Channel Estimation for Extremely Large-Scale MIMO: Far-Field or Near-Field?

Course: Principles of Wireless Communications

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1 Motivation

Extremely large-scale multiple-input multiple-output (XL-MIMO) uses hundreds of antennas to provide the array gain and spatial resolution needed for high-rate 6G links. Accurate channel state information is essential for precoding, yet the number of radio-frequency chains is much smaller than the number of antennas. Directly estimating every antenna coefficient would therefore require excessive pilot overhead.

Compressed sensing reduces this overhead by assuming that only a few paths are active in an angular Fourier dictionary. Cui and Dai show that this assumption breaks down when the XL-MIMO aperture places users or scatterers in the near field [1].

2 Far-Field and Near-Field Propagation

The boundary is approximated by the Rayleigh distance

$$Z = \frac{2D^2}{\lambda}, \tag{1}$$

where D is the array aperture and λ is the wavelength. Since Z grows quadratically with D, an XL-MIMO array can have a near-field region extending over a large part of a cell.

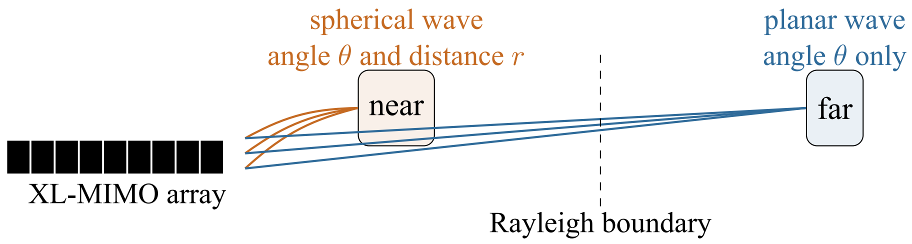

Figure 1: Propagation models emphasized in the group presentation.

Far-field phase changes linearly across the array, so one Fourier vector $\boldsymbol{a}(\theta)$ is sufficient. Near-field phase also depends on distance and requires $\boldsymbol{b}(\theta, r)$. An angle-only dictionary therefore spreads one physical near-field path over many bins.

3 Paper Contributions

The paper restores sparsity with a polar dictionary that samples angle and distance, then proposes P-SOMP for on-grid recovery and P-SIGW for off-grid refinement. The key question is which coordinates preserve physical sparsity.

4 Near-Field Channel Model

For a uniform linear array with centered antenna index n, spacing d, and wavenumber $k_c = 2\pi/\lambda$, the near-field steering vector is

$$\boldsymbol{b}(\theta, r) = \frac{1}{\sqrt{N}} \left[e^{-jk_c(r^{(0)}-r)}, \dots, e^{-jk_c(r^{(N-1)}-r)} \right]^H, \tag{2}$$

where

$$r^{(n)} = \sqrt{r^2 + n^2 d^2 - 2rnd\theta} \tag{3}$$

is the distance between the source and antenna n. Unlike a Fourier vector, Equation (2) has a nonlinear phase progression that changes with both θ and r.

With a limited number of paths, the channel remains compressible even though it is not sparse in angle. The proposed representation is

$$\boldsymbol{h}_m = \boldsymbol{W}\boldsymbol{h}_m^P, \tag{4}$$

where every column of $\boldsymbol{W}$ is a sampled steering vector $\boldsymbol{b}(\theta_p, r_p)$ and $\boldsymbol{h}_m^P$ is sparse in the polar domain. The subscript m denotes a subcarrier; the path locations are shared across subcarriers, which enables simultaneous recovery.

5 From Energy Spread to Polar Sparsity

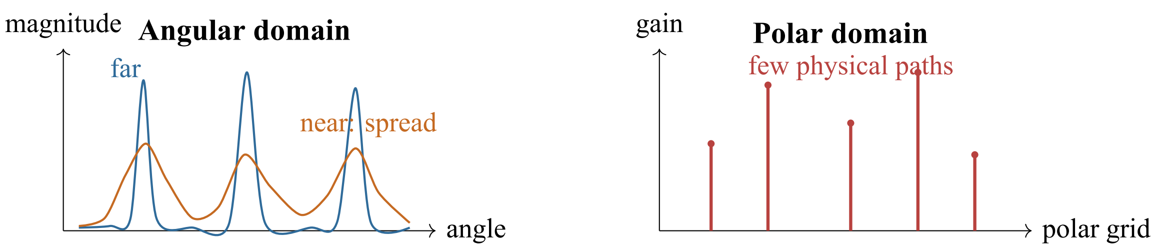

Figure 2: Energy spread and polar sparsity, adapted from Fig. 8 of [1].

Figure 2 captures the key motivation selected by the group. In the angular domain, a far-field path forms a narrow peak, while a near-field path occupies many bins. In the polar domain, each path is represented by an angle–distance pair and the channel again contains only a few significant coefficients. This restored sparsity is what makes low-pilot compressed sensing practical.

6 Dictionary Design Objective

The polar grid should not contain highly similar columns. The paper therefore designs its samples to reduce

$$\mu = \max_{p \neq q} \left| \boldsymbol{b}^H(\theta_p, r_p)\boldsymbol{b}(\theta_q, r_q) \right|, \tag{5}$$

because lower coherence improves the reliability of sparse support detection.

7 Design of the Polar-Domain Matrix W

The exact spherical distance makes direct grid design difficult. The Fresnel approximation separates its phase into

$$r^{(n)} - r \approx -nd\theta + \frac{n^2 d^2 (1-\theta^2)}{2r}. \tag{6}$$

The linear term describes angle; the quadratic term contains the coupled quantity $(1-\theta^2)/r$. This yields the two sampling rules emphasized in slides 9–17.

8 Angular Sampling on Distance Rings

Locations satisfying

$$\frac{1-\theta^2}{r} = \frac{1}{\rho} \tag{7}$$

have the same quadratic phase and lie on a *distance ring* ρ. Within one ring, coherence is governed by the same linear phase as in a far-field Fourier dictionary. Therefore the angles are sampled uniformly:

$$\theta_n = \frac{2n - N + 1}{N}, \qquad n = 0, 1, \ldots, N-1. \tag{8}$$

9 Nonuniform Distance Sampling

For vectors at the same angle, coherence depends on the difference between inverse distances. To keep adjacent columns below a chosen coherence threshold, the sampled distance at angle θ follows

$$r_s(\theta) = \frac{1}{s} Z_\Delta (1-\theta^2), \qquad s = 0, 1, 2, \ldots, \tag{9}$$

where Z_Δ is determined by the array, wavelength, and desired coherence threshold. The case $s = 0$ denotes an infinite-distance far-field ring. Thus distance is sampled uniformly in inverse distance, not uniformly in metres.

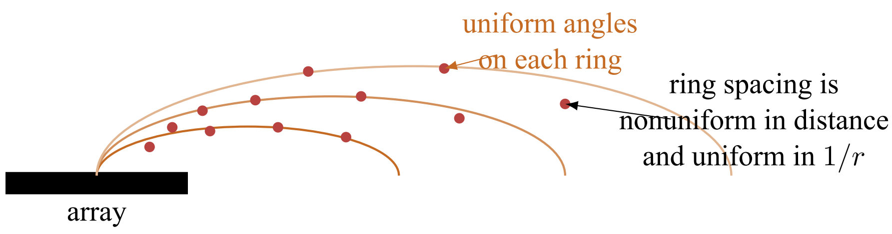

Figure 3: Conceptual distance-ring sampling used to construct $\boldsymbol{W}$.

10 Matrix Construction

Each ring produces

$$\boldsymbol{W}_s = [\boldsymbol{b}(\theta_0, r_{s,0}), \ldots, \boldsymbol{b}(\theta_{N-1}, r_{s,N-1})], \tag{10}$$

and $\boldsymbol{W} = [\boldsymbol{W}_0, \ldots, \boldsymbol{W}_{S-1}] \in \mathbb{C}^{N \times NS}$. The $s = 0$ far-field ring makes the Fourier dictionary a special case.

11 Near-Field Channel Estimation Algorithms

After analog combining, the noise is colored. The paper first applies a Cholesky-based pre-whitening transform and obtains the multi-subcarrier model

$$\overline{\boldsymbol{Y}} = \overline{\boldsymbol{\Phi}}\boldsymbol{H}^P + \overline{\boldsymbol{N}}, \qquad \overline{\boldsymbol{\Phi}} = \boldsymbol{D}^{-1}\boldsymbol{A}\boldsymbol{W}. \tag{11}$$

The rows of $\boldsymbol{H}^P$ share a common sparse support because path angles and distances are common across subcarriers.

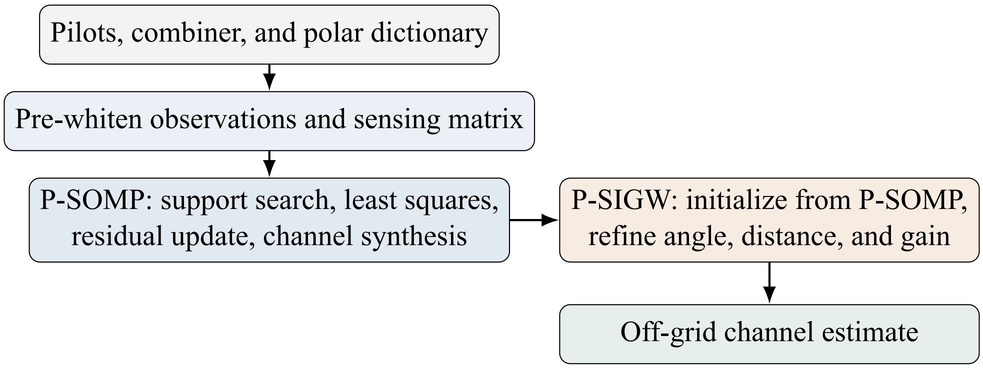

Figure 4: Combined workflow of the two algorithms emphasized in the group presentation.

11.1 On-Grid P-SOMP

P-SOMP extends simultaneous OMP from the angular dictionary to $\boldsymbol{W}$. It proceeds as follows:

1. construct $\boldsymbol{W}$ and pre-whiten the measurements;
2. initialize the residual and an empty support set;
3. select the polar atom with the largest summed correlation across subcarriers;
4. estimate gains by least squares and update the residual;
5. repeat for the detected paths and synthesize the channel.

Because $s = 0$ represents infinite distance, P-SOMP also works for far-field paths. Its weakness is grid mismatch: real angles and distances are continuous and rarely coincide exactly with sampled atoms.

11.2 Off-Grid P-SIGW

P-SIGW uses P-SOMP only for initialization. It constructs a small steering matrix from the detected paths and minimizes the whitened reconstruction error

$$\min_{\boldsymbol{\theta},\boldsymbol{r},\boldsymbol{G}} \left\| \overline{\boldsymbol{Y}} - \widetilde{\boldsymbol{\Phi}}(\boldsymbol{\theta}, \boldsymbol{r})\boldsymbol{G} \right\|_F^2 . \tag{12}$$

Alternating updates refine angle and inverse distance, followed by a least-squares gain update. This directly estimates continuous path parameters and removes the resolution floor of P-SOMP.

12 Trade-Off

P-SOMP is simpler but bounded by grid resolution. P-SIGW corrects off-grid mismatch and improves NMSE at the cost of iterative optimization.

13 Simulation Results

The distance experiment uses $N = 256$ antennas, a 100 GHz carrier, SNR $= 10$ dB, and pilot length $P = 32$. The corresponding Rayleigh distance is about 100 m. Figure 5 redraws the principal trends from Fig. 10 of the paper rather than reproducing its image.

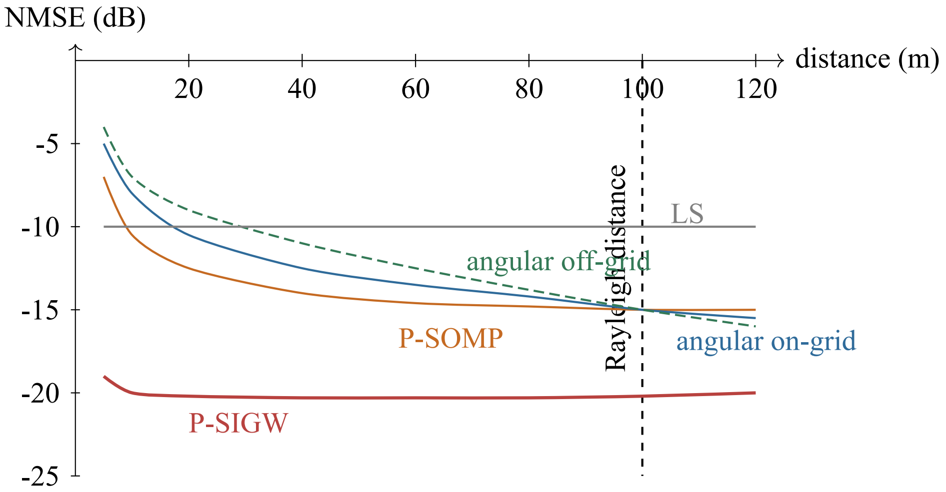

Figure 5: Trend-focused redraw of NMSE versus distance from [1, Fig. 10].

14 Interpretation

Inside the Rayleigh distance, angular-domain methods suffer severe model mismatch because a spherical path is not sparse in angle. Their NMSE improves only as distance increases and the wavefront becomes nearly planar. P-SOMP is much less sensitive to distance because its dictionary contains the required angle–distance atoms. P-SIGW is best because it also removes the polar-grid quantization error.

The other experiments selected by the group support the same conclusion. As SNR increases, near-field polar methods continue improving while mismatched angular methods approach an error floor. Increasing pilot length improves all compressed-sensing estimators, but the gain is much larger when the representation matches the propagation physics.

15 Conclusion

XL-MIMO does not destroy physical channel sparsity; it destroys *angle-only* sparsity. Uniform angle sampling on distance rings and nonuniform sampling in $1/r$ let $\boldsymbol{W}$ concentrate spherical paths. P-SOMP exploits this structure with low pilot overhead, while P-SIGW trades extra computation for off-grid accuracy.

References

[1] M. Cui and L. Dai, "Channel estimation for extremely large-scale MIMO: Far-field or near-field?" *IEEE Trans. Commun.*, vol. 70, no. 4, pp. 2663–2677, Apr. 2022, doi: 10.1109/TCOMM.2022.3146400.

Q&A

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